Positional Encoding in Transformers

CS115 – Spring 2026

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Today's plan

- Why position matters at all
- Absolute positional encoding
- Relative position methods
- RoPE and why modern LLMs love it
- ALiBi and long-context intuition
- Comparison, design choices, quiz

Big question

How can a Transformer know that

`dog bites man`

is not the same as

`man bites dog?`

Learning objectives

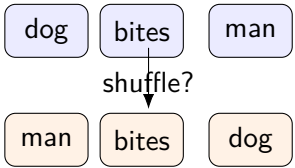
By the end of this lecture, you should be able to:

1. explain why self-attention needs explicit positional information,
2. describe **absolute** vs. **relative** positional encoding,
3. interpret the original **sinusoidal** encoding geometrically,
4. explain how **RoPE** turns position into rotations,
5. explain how **ALiBi** adds a distance-dependent attention bias,
6. compare trade-offs for generalization, extrapolation, and efficiency.

Without position, attention is too symmetric

- A Transformer layer sees a *set* of token embeddings.
- Self-attention can learn *relationships*, but if we never tell it where tokens are, it has no built-in notion of order.
- In that sense, pure attention is close to being permutation-equivariant.

Informal consequence: the model can know *what* words are present, but not reliably *where* they occur.



A toy failure case

Same bag of words, different meaning

`the cat chased the mouse` `the mouse chased the cat`

- Word identities alone do not tell us the agent, the patient, or the direction of an event.
- Sequence models need to represent at least one of these:
  - the **absolute location** of a token, or
  - the **relative distance** between pairs of tokens.

What should a positional scheme do?

A useful positional encoding should ideally help with:

Local structure

- adjacent tokens
- short-range syntax
- phrase boundaries

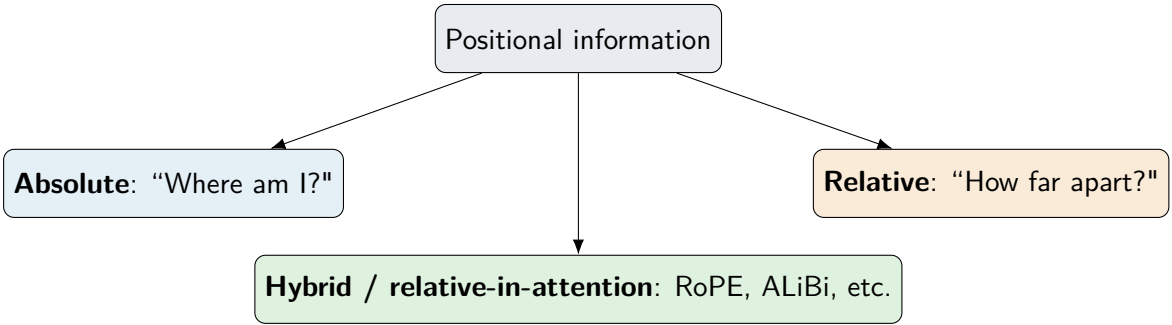
Global structure

- far-apart references
- document layout
- long-context reasoning

Engineering wish list

Generalize beyond training length, preserve efficiency, and play nicely with attention.

Taxonomy: three ways to think about position



- **Absolute**: assign each token a location code.
- **Relative**: modify attention based on offsets between tokens.
- **Hybrid**: encode absolute positions so that *attention interactions* depend on relative displacement.

Metaphor: addresses vs. distances

Absolute position

Like a **street address**:

- token x_5 is at house #5,
- token x_{12} is at house #12.

Relative position

Like asking:

- how many houses apart are we?
- am I 1 step left, 3 steps right, 20 steps away?

Rule of thumb

Absolute position is good for *location*. Relative position is good for *relationships*.

Why sine and cosine? A bank of clocks

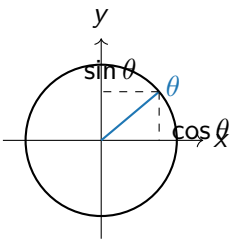
- Imagine every token position is observed by many clocks.
- Some clocks spin fast, some spin slowly.
- Position p becomes a collection of clock-hand coordinates.



Key metaphor:

A single clock repeats too often. Many clocks with different speeds together can identify position surprisingly well.

Unit-circle intuition: one coordinate pair = one little compass



- A point on the unit circle is determined by angle θ.
- Its coordinates are $(\cos\theta, \sin\theta)$.
- So a 2D pair can encode “where the hand is pointing."

For positional encoding, we let the angle depend on the token position.

Original Transformer sinusoidal encoding

For model dimension d_{model}:

$$\mathrm{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right),$$
$$\mathrm{PE}(\text{pos}, 2i+1) = \cos\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right).$$

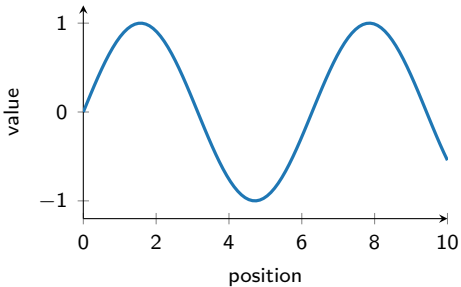
- Dimensions come in **pairs**: one sine, one cosine.
- Each pair uses a different wavelength / frequency.
- Low-index dimensions vary quickly; high-index dimensions vary slowly.

Interpretation

Each 2D pair acts like a clock rotating at its own speed.

One frequency alone is not enough

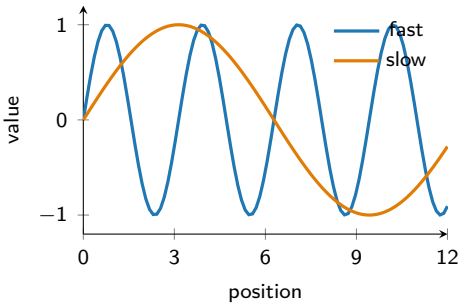
- A single sine wave repeats periodically.
- If we only used one frequency, many positions would look identical.
- So we combine many frequencies.



Many clocks together reduce ambiguity.

Why many frequencies help

- Fast waves capture fine local distinctions.
- Slow waves capture coarse global location.
- Together they produce a multi-scale signature for each position.



Analogy

Like reading time from several clocks geared differently: one tells seconds, another minutes, another hours.

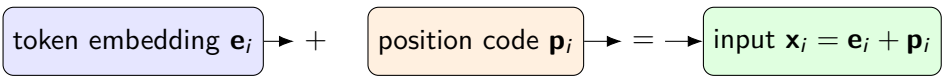
Tiny numerical example

Suppose the model uses only 4 positional dimensions (2 sine/cosine pairs). Then position codes might look like:

position	dim 0	dim 1	dim 2	dim 3
0	$\sin 0 = 0$	$\cos 0 = 1$	$\sin 0 = 0$	$\cos 0 = 1$
1	$\sin(1)$	$\cos(1)$	$\sin(0.01)$	$\cos(0.01)$
2	$\sin(2)$	$\cos(2)$	$\sin(0.02)$	$\cos(0.02)$
3	$\sin(3)$	$\cos(3)$	$\sin(0.03)$	$\cos(0.03)$

- The first pair changes quickly.
- The second pair changes slowly.
- Position becomes a **vector signature**, not a single number.

How absolute positional encoding enters the model



- This is the classic **absolute** approach.
- The position vector is injected before attention.
- The model must learn to use this mixed signal effectively.

Learned absolute embeddings

Instead of fixed sine/cosine vectors, we can learn a table:

position 0	learned vector $\mathbf{p}_0$
position 1	learned vector $\mathbf{p}_1$
position 2	learned vector $\mathbf{p}_2$
$\vdots$	$\vdots$
position $L_{\max}$	learned vector $\mathbf{p}_{L_{\max}}$

- Very common in early Transformer NLP models.
- The network learns whichever codes help the training objective.
- But it usually ties you to a maximum trained length.

Examples of absolute schemes in classic NLP models

Fixed absolute

- Original Transformer (Vaswani et al.)
- Sinusoidal encoding

Learned absolute

- BERT-style learned position embeddings
- GPT-2-style learned position embeddings

Shared strength

They directly tell the model *where* each token sits.

Strengths of absolute positional encoding

- Easy to understand and easy to implement.
- Gives every position an explicit identity.
- Fixed sinusoidal encoding adds **no learned parameters**.
- Sinusoidal patterns can generalize somewhat beyond training length better than naive learned tables.

Best mental model

Absolute encoding answers: **"What is my address in the sequence?"**

Limits of absolute positional encoding

- Attention often cares more about **relative distance** than raw address.
- Learned position tables usually do not extrapolate gracefully to much longer sequences.
- Even fixed sinusoidal encoding does not automatically make pairwise relations easy to learn.
- Adding position to embeddings can mix content and position in a somewhat indirect way.

Transition

This motivates methods that inject position directly into the *attention score* between token pairs.

Why relative position feels natural for attention

When token i attends to token j, the important question is often:

$$\text{How far away is } j \text{ from } i?$$

Examples:

- A word often depends on the token immediately before it.
- A closing bracket may depend on a matching opening bracket many steps earlier.
- Pronouns may link to nouns a few or many tokens away.

Relative offsets form a pairwise matrix

For a sequence of 5 tokens, the relative offset $j - i$ looks like:

$i \backslash j$	1	2	3	4	5
1	0	1	2	3	4
2	-1	0	1	2	3
3	-2	-1	0	1	2
4	-3	-2	-1	0	1
5	-4	-3	-2	-1	0

- Positive: key is to the right.
- Negative: key is to the left.
- Magnitude tells us distance.

Shaw-style relative position: add offset information to attention

A simple relative attention idea is:

$$\text{score}(i, j) = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d}} + b_{j-i}$$

where b_{j-i} is a learned bias for the relative offset.

- If j is 1 step away, use one bias.
- If j is 5 steps away, use another bias.
- Now the attention score explicitly depends on relative distance.

Relative position can also use vectors, not just scalars

More expressive variants use relative-position *embeddings*:

$$\text{score}(i, j) = \frac{\mathbf{q}_i^\top (\mathbf{k}_j + \mathbf{r}_{j-i})}{\sqrt{d}}$$

- $\mathbf{r}_{j-i}$ is a vector for offset $(j - i)$.
- This lets the model learn different geometric patterns for different distances.
- But memory and implementation can become more complicated.

T5-style bucketed relative bias

T5 uses a smart compromise:

$$\text{score}(i,j) = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d}} + b_{\text{bucket}(j-i)}$$

- Nearby offsets get fine-grained buckets.
- Far-away offsets get coarser buckets.
- So the model distinguishes local distances precisely without needing one parameter for every possible offset.

Good intuition

T5 says: *"I care exactly whether you are 1 or 2 words away, but for very far positions I only need rough distance bins."*

Examples of relative methods

- **Shaw et al.**: learned relative embeddings/biases in attention.
- **T5**: bucketed relative attention bias.
- **DeBERTa**: disentangles content and position interactions.
- **RoPE**: rotate queries/keys so dot products depend on relative offsets.
- **ALiBi**: add a simple linear distance penalty to attention scores.

Notice

Some methods are *purely relative*; some are hybrids that start from absolute positions but produce relative effects inside attention.

Strengths and limits of relative approaches

Strengths

- attention gets distance info directly
- often better inductive bias for language
- translation-like invariance: same pattern can work in different locations

Limits

- some forms are harder to implement efficiently
- some need clipping or bucketing
- long-range extrapolation still depends on design details

RoPE: the most influential modern idea

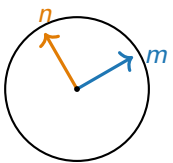
Big picture

Instead of adding a position vector to the token embedding, **RoPE rotates query and key vectors by an angle that depends on position.**

- Position m means “rotate by angle proportional to m."
- Different dimension pairs rotate at different rates.
- The magic: when rotated queries and keys take a dot product, the result depends on the **relative angle difference**.

RoPE intuition: every query/key carries a rotating compass

- Each 2D slice of a vector acts like a little arrow in the plane.
- At position m, rotate that arrow by angle $m\theta$.
- At position n, rotate by $n\theta$.
- Comparing the two arrows naturally reveals the offset $n - m$.



angle gap encodes offset

RoPE math in one 2D pair

For a 2D vector slice $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ at position m:

$$R(m\theta) = \begin{bmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{bmatrix}$$

and we use

$$\mathbf{q}'_m = R(m\theta)\mathbf{q}_m, \qquad \mathbf{k}'_n = R(n\theta)\mathbf{k}_n.$$

Exactly like sinusoidal encoding, different 2D pairs use different frequencies θ_i.

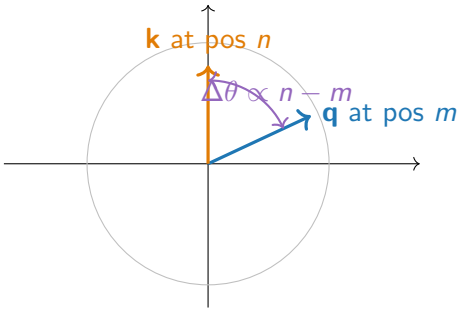
Why RoPE becomes relative inside attention

The key identity is:

$$(R(m\theta)\mathbf{q})^\top (R(n\theta)\mathbf{k}) = \mathbf{q}^\top R((n-m)\theta)\mathbf{k}.$$

- The dot product depends on $n - m$, not separately on m and n.
- So RoPE injects position in a way that is naturally compatible with relative attention.
- This is why RoPE is often described as **absolute input, relative effect**.

Geometric picture of RoPE



Metaphor

RoPE turns position into **phase**. Attention then compares **phase differences**.

Tiny worked RoPE example

Take one 2D query slice $\mathbf{q} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and one key slice $\mathbf{k} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

- At position $m = 1$: $R(\theta)\mathbf{q} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$
- At position $n = 4$: $R(4\theta)\mathbf{k} = \begin{bmatrix} \cos 4\theta \\ \sin 4\theta \end{bmatrix}$

Their dot product is:

$$\cos\theta\cos 4\theta + \sin\theta\sin 4\theta = \cos(3\theta).$$

Only the relative gap $4 - 1 = 3$ remains.

Why RoPE became popular in LLMs

- Gives a strong relative inductive bias without huge attention-side tables.
- Works naturally in decoder-only causal self-attention.
- Parameter-free as a positional mechanism.
- Often performs very well in practice for language modeling and long-context tasks.
- Easy to combine with context-extension tricks (scaled RoPE, YaRN, interpolation, etc.).

RoPE in modern open-weight LLMs

Examples from recent model documentation / reports

- **Llama 3.x**: official/open configs expose `rope_theta` and scaled RoPE settings.
- **DeepSeek-V3**: applies RoPE to the query/key components inside MLA.
- **Qwen2.5 / Qwen2.5-1M**: explicitly discuss RoPE-based models and RoPE-frequency scaling for long context.

Evidence examples: Meta Llama 3 code/configs list `rope_theta`; DeepSeek-V3 describes keys/queries carrying Rotary Positional Embedding; Qwen2.5-1M reports "Modern LLMs based on RoPE" and adjusts RoPE base frequency for longer contexts.

RoPE is powerful, but long context is tricky

- RoPE uses angles that grow with position.
- When we go far beyond training length, the model sees phase patterns it never learned to use.
- Result: performance can degrade for very long sequences.

Intuition

If the clock hands spin into unfamiliar regions, the model may misread the geometry.

Modern fixes for long-context RoPE

Common strategies:

- **Scaled RoPE / position interpolation**: compress positions so long contexts map into a familiar phase range.
- **YaRN**: adjusts scaling so long sequences are easier to handle without destroying short-range behavior.
- **ABF (Adaptive Base Frequency)**: change the RoPE base frequency during long-context training.
- **DCA (Dual Chunk Attention)**: remap relative distances so they stay within trained ranges for very long inputs.

Current trend

Many SOTA/open LLMs keep **RoPE** but modify how position is scaled or how long-range attention is organized.

ALiBi: Attention with Linear Biases

Core idea

Do not rotate vectors. Do not add position embeddings.
Just add a **distance-dependent linear penalty** to attention scores.

- Nearby tokens get less penalty.
- Faraway tokens get more penalty.
- Different heads can have different slopes, so some heads stay local and others see farther.

ALiBi formula

In causal attention, for token i attending to past token $j \leq i$:

$$\text{score}(i, j) = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d}} - m_h(i - j)$$

where:

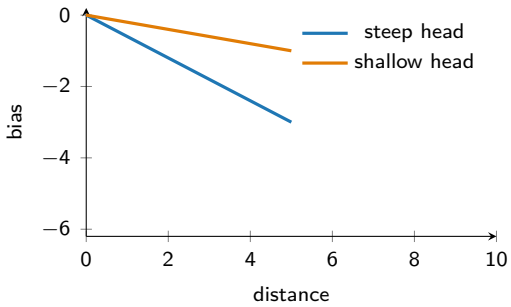
- $m_h > 0$ is the slope for head h,
- $(i - j)$ is the distance into the past.

Interpretation

ALiBi adds a **distance tax** to attention.

ALiBi intuition: some heads are local, some are global

- A steep slope means “prefer close tokens strongly.”
- A shallow slope means “long-range attention is still affordable.”
- Multiple heads create multiple distance preferences.



Why ALiBi is attractive

- Extremely simple.
- No learned position table.
- Encourages length extrapolation better than many naive absolute schemes.
- Keeps attention score computation conceptually clean.

Mental model

ALiBi does not say *where* a token is. It says how much the model should *discount* it because of distance.

ALiBi vs. RoPE

RoPE

- rotates queries/keys
- phase-based relative effect
- widely used in todayś open LLMs
- strong practical performance

ALiBi

- adds linear distance bias
- extremely simple
- often strong for extrapolation
- may be less expressive geometrically

Comparison table

Method	Injects what?	Where injected?	Strength	Weakness
Sinusoidal absolute	absolute location	add to embeddings	simple; no learned parameters	indirect for pairwise relations
Learned absolute	absolute location	add to embeddings	flexible and fully learnable	weak extrapolation beyond training length
Relative bias (Shaw/T5)	relative offset	attention score	natural inductive bias for attention	may need buckets, clipping, or extra tables
RoPE	absolute phase with relative interaction	rotate Q, K	elegant, strong, widely used	long-context scaling needs care
ALiBi	linear distance penalty	attention score	very simple; good extrapolation intuition	less expressive than rotation-based geometry

How to choose in practice

- If you want the most historically standard modern decoder-only LLM choice: **RoPE**.
- If you want very simple long-range biasing: **ALiBi**.
- If you want explicit learned relative attention in encoder/encoder-decoder settings: **T5-style relative bias** is natural.
- If you want basic pedagogy or a small baseline: **sinusoidal absolute** is still excellent for teaching.

Important

There is no single universally best positional encoding. The best choice depends on architecture, training length, target context window, and implementation constraints.

Common misconceptions

1. **"Attention already knows order."** No. Not unless order information is injected somehow.
2. **"Absolute and relative are mutually exclusive."** Not always. RoPE is a classic bridge.
3. **"Sin/cos is arbitrary magic."** No. It is a structured multi-frequency coordinate system.
4. **"Longer context just means bigger position numbers."** Not quite. The model must *generalize* to those new positional patterns.
5. **"RoPE solves long context for free."** No. In practice people often rescale or adapt it.

Quiz 1

Think-pair-share

Why is a **single** sine wave not enough to encode position uniquely over long sequences?

1. Because sine values are too small.
2. Because sine repeats periodically, so different positions can look the same.
3. Because sine cannot be differentiated.
4. Because attention cannot use continuous values.

Quiz 1 answer

Correct answer: 2

A single sine wave repeats. So many positions would share the same value. Using **many sine/cosine pairs at different frequencies** creates a richer position signature.

Takeaway: sinusoidal encoding is a *multi-scale coordinate system*, not just one wave.

Quiz 2

Think-pair-share

Which statement best captures why RoPE is often called a *relative-friendly* positional method?

1. It removes attention entirely.
2. It stores one embedding vector for each relative offset.
3. Rotated query-key dot products depend on the **difference** in positions.
4. It always extrapolates perfectly to any context length.

Quiz 2 answer

Correct answer: 3

Because

$$(R(m\theta)\mathbf{q})^\top (R(n\theta)\mathbf{k}) = \mathbf{q}^\top R((n-m)\theta)\mathbf{k},$$

the interaction depends on the relative offset $n - m$.

One-slide summary

- Transformers need positional information because attention alone does not encode order.
- **Absolute** methods answer: *Where is this token?*
- **Relative** methods answer: *How far apart are these tokens?*
- Sinusoidal encoding uses **many clocks** at different speeds.
- **RoPE** rotates queries/keys so attention depends on phase differences.
- **ALiBi** adds a simple distance tax to attention scores.
- Modern LLMs commonly use **RoPE or RoPE-style scaling tricks** for strong practical performance.

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Attention Is All You Need. NeurIPS 2017.


P. Shaw, J. Uszkoreit, A. Vaswani.
Self-Attention with Relative Position Representations. NAACL 2018.


C. Raffel et al.
Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. JMLR 2020.


O. Press, N. Smith, M. Lewis.
Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation. 2021.

References II



J. Su et al.

RoFormer: Enhanced Transformer with Rotary Position Embedding. 2021 / Neurocomputing 2024.



Meta AI.

The Llama 3 Herd of Models. 2024, plus official Llama 3 code/configs exposing RoPE parameters.



DeepSeek-AI.

DeepSeek-V3 Technical Report. 2024. Uses RoPE within MLA query/key components.



Qwen Team.

Qwen2.5 Technical Report. 2024 and *Qwen2.5-1M Technical Report*. 2025. Discuss RoPE-based models, ABF, DCA, and YaRN for long context.

For an undergraduate class, these are the main papers/reports worth reading next.